Vinayak Kamat

Junior AI/ML Engineer

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vinayakkamatcodes

Summary

Junior AI/ML Engineer at KAN Innovations building production-ready AI systems. Experience with end-to-end RAG pipelines, PyTorch-to-ONNX inference optimization, and ML services on AWS (Lambda, ECR, DynamoDB). Built on-device AI with llama.cpp and a real-time hallucination detection gateway. Pursuing an MSc in Financial Engineering.

Skills

Languages: Python, C++, Dart, JavaScript

ML & AI: PyTorch, Transformers, OpenCV, spaCy

LLM & RAG: LangChain, LlamaIndex, GGUF models, prompt engineering, Chroma

MLOps: MLflow, model versioning, experiment tracking

Backend & DevOps: FastAPI, Flask, Docker, AWS (Lambda, S3, ECR, DynamoDB)

Data: Pandas, NumPy

Experience

KAN Innovations, Junior AI/ML Engineer — Mumbai, Maharashtra
May 2025 – present
1 year 6 months

- Improved model calibration to enhance reliability of ML predictions in production
- Optimized inference by converting PyTorch models to ONNX for faster runtime performance
- Integrated ML models into production pipelines using AWS (Lambda, ECR, S3, DynamoDB)
- Built and evaluated Retrieval-Augmented Generation (RAG) pipelines
- Used MLflow for experiment tracking, model versioning, and reproducible workflows

KAN Innovations, ML Engineer Intern — Mumbai, Maharashtra
Feb 2025 – May 2025
3 months

- Built a PyTorch-based ulcer detection model achieving ~85% accuracy
- Improved performance of an eye posture detection model

Projects

Aethelgard

AI governance gateway for real-time analysis of LLM outputs to detect hallucinations and risk factors.

- Built a real-time LLM response analyzer using a secondary "surgeon" model for semantic risk assessment
- Implemented a FastAPI gateway streaming risk metrics to Datadog
- Designed automated hallucination detection logic for safer AI deployments

nanoMind

On-device AI chat app running LLMs completely offline using llama.cpp and Kotlin.

- Enabled offline GGUF model inference on Android devices
- Implemented real-time token streaming using Kotlin Coroutines and Jetpack Compose
- Optimized LLM loading with ARM-specific performance improvements

Financial AI Agent (Demo)
Prototype AI agent for financial data retrieval and reasoning using Python and language models.

- Built retrieval and reasoning pipelines for financial insights
- Implemented modular Python components for query routing and response refinement

Education

MSc **WorldQuant University**, Financial Engineering — Remote, Washington DC
Sept 2025 – May 2027

B.Tech **Punjab Technical University**, Computer Science — Punjab, India
Sept 2021 – May 2025

Achievements

Qualified the Graduate Aptitude Test in Engineering (GATE)

Participated in Kaggle competitions focused on applied machine learning

Active open-source contributor with AI/ML projects on GitHub